

Assignment 3 — Reflection

CART 498 · GPT-4.1, the Worst Mathematician Ever · Dan Truong

For this assignment I made a bot that asks GPT-4.1-nano to do multiplication for me. It takes a number, multiplies it by itself, then does it again with the answer, a few times in a row. After each step, Python checks if GPT got it right. If it's wrong, the bot makes fun of itself.

Honestly I thought an AI would be really good at math, because a calculator can do this in a second. But the bot broke on 7 out of the 11 number combinations I tried. With small numbers it was always right. Once the answers got really long, like more than 10 digits, it started getting most of them wrong. One time it said $15129 \times 15129 = 2286142441$, but the real answer is 228886641. It even added an extra number.

From what I understand, GPT doesn't actually calculate like a calculator does. It's more like it guesses what the answer should look like, based on all the text it learned from. It also reads long numbers in small pieces instead of as one whole number. So the beginning of the answer usually looks right and then the end goes wrong. For one answer the first 9 digits were correct and the rest was just random. Another time it wrote "174174174" over and over, and once it just gave up and said "948,000,000,000", like a rough guess.

I also told it to only give me the number with no explanation. I think that made it harder, because when I do math on paper I need to write the steps down, and the bot can't do that.

The weird part is that it got some big numbers right, like 4294967296. I looked it up and it's a famous number in computers, so I think GPT just remembered it instead of really solving it.

The funny thing is my bot never got really sad. It only got up to "frustrated", because it never made more than 2 mistakes in one try. I made a whole existential crisis mood for nothing.